# Exp 1a – Making Measurements

## INTRODUCTION

Mass, volume, and length are fundamental measurements made in any laboratory. In the following experiment, you will be calculating the density of a metal bar based on mass, volume, and length measurements. You will also determine the density of pure aluminum shot and determine whether the metal bar is made of aluminum or some other metal.

## BACKGROUND

Measuring Volume

In laboratories, a common device for measuring volumes of liquids is the graduated cylinder (see figure 1). Although it is only used for measuring liquid volumes, it is often used to indirectly measure the volume of solid objects via the "displacement method" discussed below. The volume measured is based on the lowest point of the meniscus; the curved shape created by the liquid in the graduated cylinder. Scientists always estimate one digit beyond the markings shown on the measuring device. The graduated cylinder to the right has markings for every 1 mL, so measurements should always be recorded to the nearest 0.1 mL. When the meniscus is exactly on the measurement line as shown in the picture to the right, the decimal recorded would be zero. Using this method, we would record that the volume of water in the cylinder is 43.0 mL.

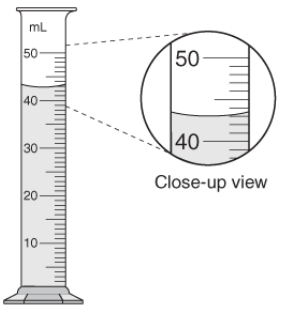



**Figure 1: A 50-mL Graduated Cylinder**

The volume of a geometrically regular shaped solid can be determined by first measuring its dimensions and then calculating its volume using the volume equation for that shape. For example, consider a rectangular prism (see figure 2). The volume can be determined by measuring the width, the length, and the height. The volume can then be calculated by:

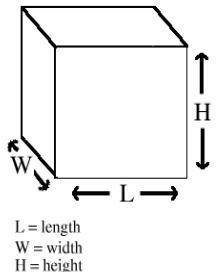



**Figure 2: The Volume of a Rectangular Prism**

$$V_{reactangular\ prism} = width \times length \times height \quad (1)$$

For a cylinder, the following volume equation is used:

$$V_{cylinder} = \pi \times radius^2 \times height \quad (2)$$

The volume of a regularly shaped and an irregularly shaped object can be determined by *displacement*. A volume of liquid is measured first without an object and then is measured again with the object. For example, if you wanted the volume of an egg, you could first put one cup of water in a measuring cup and then add the egg (fresh eggs sink). The additional measured volume is the volume of the egg. In general, the displaced volume can be calculated by:

$$Egg\ Volume = Final\ Volume - Initial\ Volume \quad (3)$$

When an object is irregularly shaped (i.e., the dimensions can't easily be measured), it is usually easiest to measure the volume by displacement.

Measuring Mass

A "balance" is used for measuring the mass of an object. The term balance is an archaic term that comes from the days when mass was measured by balancing masses (see figure 3). Modern balances measure the force exerted by an object and compare it to the force created by an object of known mass (calibration to a standard).



**Figure 3: A Balance**

A simple method for determining the mass of an object with an analytical balance (see figure 4) is as follows:

1. Tare the balance by hitting the "0" or tare button.
   * If necessary, place weighing paper on the balance to protect the metal balance pan from the substance you are measuring.
   * Make sure the balance doors are shut, and you are not leaning on the counter.
   * Push the gray bar if the digital readout does not show 0.0000 g.
2. Place the object to be massed on the balance pan.
   * Be sure that the object is clean.
   * Be sure to reclose the doors after adding the object.



**Figure 4: An Analytical Balance**

3. When the display stabilizes, record the object's mass. (Always record all decimals shown on the balance.)

If you haven't recognized it, this is the same process they use at the meat counter in the grocery store ☺.

**ADDITIONAL READING**

- Measurements, Problem Solving, Unit Conversions and Density

**SAFETY EQUIPMENT OPTIONAL FOR LAB**

- Goggles
- Lab coat

**SUPPLY LIST**

These are items that you can gather before beginning your experiment.

- Metal bar
- Ruler
- 50 or 100-mL graduated cylinder
- Glass stirring rod
- Wash bottle with distilled water

**CLASSROOM SUPPLIES AND REACTANTS**

These are shared supplies or reactants. Do not gather these items. Leave these items at their designated location in the classroom and use as needed according to the procedure.

- Aluminum shot
  (shot = pellets)

**PROCEDURE**

* Record all qualitative and quantitative data *neatly* on the "Data and Report Pages". Remember to include both units and correct significant figures.

1. ☐ Record the unknown number or letter of your metal bar. Briefly describe its appearance.
2. ☐ Measure the mass of the metal bar using an analytical balance.

* Record all the digits visible on the balance display. (*Always record all the digits* when reading a balance.

3. ☐ Measure the dimensions (l, w, h or diameter) of the metal bar in cm units.

   * Record these values to the correct number of significant figures (remember estimate 1 place past the smallest division).

4. ☐ Measure the volume of the metal bar by displacement using water and a graduated cylinder.
   a. Record the volume of water before adding the bar (remember, *always* estimate 1 place past the smallest division.)
   b. Add the metal bar very carefully by first tilting the graduated cylinder slightly to one side, then by allowing the metal bar to slide slowly to the bottom.

      * Check for air bubbles. Remove any large bubbles with the glass rod.

   c. Record the volume of water after adding the bar.
   d. Pour the water out of the graduated cylinder and dry the metal bar with a paper towel.
5. ☐ Obtain a small amount of dry aluminum shot (between 10 g to 30 g is fine) and measure its mass. Record the precise mass on your report sheet.
6. ☐ Briefly describe its appearance on your report sheet.
7. ☐ Measure the volume of the metal shot by displacement.

   * Be sure to remove air bubbles using the glass rod.

**CLEAN UP AND WASTE DISPOSAL**

1. ☐ After checking your data with your instructor, dry the shot with a paper towel before putting them in the "return metal" container.
2. ☐ Dry your metal bar with a paper towel before putting it back in the original bin.
3. ☐ Empty your water bottle before putting it back.
4. ☐ Put all other equipment back where you found it.*

   *There is no need to thoroughly dry the graduated cylinder prior to returning it to the shelf. Never attempt to dry the inside of a graduated cylinder with a paper towel!! More often than not, the paper towel gets stuck inside.

**DATA AND REPORT – DUE ____ / ____ / ____** **SECTION _____**

* Show your work and keep track of significant figures. **DATE _____**

**PARTNER _____**

**DATA**

1. Number or letter on the bar: _____
2. Mass and appearance of the metal bar.
   a. Mass _____
   b. Shape _____
   c. Color _____
3. Dimensions of the bar:
   a. Longest side (Length) – or Diameter (for cylinders): _____
   b. Medium side (Height): _____
   c. Short side (Width): _____ (not needed for cylinders)
4. Water displacement of the bar:
   a. Initial Water Volume _____
   b. Final Water Volume _____
5. Mass and appearance of the aluminum shot:
   a. Mass _____
   b. Shape _____
   c. Color _____
6. Water displacement of the aluminum shot:
   a. Initial Water Volume _____
   b. Final Water Volume _____

**RESULTS**

* Show your work and keep track of significant figures.

7. Calculate the metal bar's volume (in $cm^3$) using its dimensions.

8. Calculate the volume of the bar in mL using water displacement.

9. Chemists compare two values mathematically by calculating percent difference. Calculate the percent difference between the two volumes using the following equation:

$$\%\ Difference = \frac{Volume\ from\ dimensions - Volume\ by\ displacement}{Volume\ by\ displacement} \times 100$$

10. How does the bar's 'volume measured by dimensions' compare to the bar's 'volume measured by displacement'? (Use complete sentences, mentioning both numerical volume measurements, and the percent difference)

11. Calculate the density of the metal bar. (Use the volume from the displacement measurement.)

12. Determine the density of the aluminum shot.

13. Compare the densities of the aluminum shot to the density of the metal bar you measured earlier. Could your bar be made of aluminum? Why or why not? If the bar is not aluminum, what could the metal be?

| **Metal** | **Density ($g/cm^3$)** |
|---|---|
| Brass | 8.5 |
| Cast Iron | 6.7 |
| Copper | 9.0 |
| Gold | 19.3 |
| Lead | 11.3 |
| Silver | 10.5 |
| Stainless Steel | 7.5 |
| Steel | 8.0 |
| Titanium | 4.5 |
| Tin | 5.8 |